

**B.Tech. Degree I Semester Regular Examination November 2023****CE/ME/SE 23-200-0101A & EE/EC/CS/IT 23-200-0101B CALCULUS**  
(2023 Scheme)

Time: 3 Hours

Maximum Marks: 50

**Course Outcomes**

On successful completion of the course, the students will be able to:

- CO1: Solve ordinary differential equations and linear differential equations of higher orders with constant coefficient and apply them in engineering problems.  
 CO2: Determine the maxima and minima of multi variable functions.  
 CO3: Evaluate area as double integrals and volume as triple integrals.  
 CO4: Illustrate the application and physical meaning of gradient, divergence and curl.  
 Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create  
 PO – Programme Outcome

**PART A**

(Answer ALL questions)

(5 × 2 = 10)

|                                                                                               | Marks | BL | CO | PO |
|-----------------------------------------------------------------------------------------------|-------|----|----|----|
| I. (a) Check whether $(1-x)dx - (1+y)dy$ is exact or not.                                     | 2     | L1 | 1  | 1  |
| (b) If $x^y = y^x$ , find $\frac{dy}{dx}$ .                                                   | 2     | L1 | 2  | 1  |
| (c) Evaluate $\int_0^2 \int_1^2 (x^2 + y^2) dx dy$ .                                          | 2     | L2 | 3  | 1  |
| (d) Compute the divergence of $\vec{F} = xyz\hat{i} + xyz\hat{j} + xyz\hat{k}$ at $(1,1,1)$ . | 2     | L2 | 4  | 1  |
| (e) Show that $\vec{F} = (x+3y)\hat{i} + (y-2z)\hat{j} + (x-2z)\hat{k}$ is solenoidal.        | 2     | L1 | 4  | 1  |

**PART B**

(4 × 10 = 40)

|                                                                                     |   |    |   |   |
|-------------------------------------------------------------------------------------|---|----|---|---|
| II. (a) Solve $(D^2 - 2D + 1)y = xe^x \sin x$ .                                     | 5 | L3 | 1 | 1 |
| (b) Solve the system $\frac{dx}{dt} = 2y, \frac{dy}{dt} = 2z, \frac{dz}{dt} = 2x$ . | 5 | L6 | 1 | 1 |

**OR**

|                                                                                                                    |   |    |   |   |
|--------------------------------------------------------------------------------------------------------------------|---|----|---|---|
| III. (a) Solve $(D^2 + 4D + 4)y = \cos 5x + e^{3x}$ .                                                              | 5 | L5 | 1 | 1 |
| (b) Solve $(x^2 D^2 - 2xD + 2)y = x^3$ .                                                                           | 5 | L4 | 1 | 1 |
| IV. (a) Verify Euler's Theorem for the function $u = ax^2 + 2hxy + by^2$ .                                         | 5 | L3 | 2 | 1 |
| (b) If $x = r \cos t$ and $y = r \sin t$ . Find $J = \frac{\partial(x,y)}{\partial(r,t)}$ . Show that $JJ^I = 1$ . | 5 | L5 | 2 | 1 |

**OR**

|                                                                                                                                                                 |   |    |   |   |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----|---|---|
| V. (a) Find the maximum and minimum values of $f(x,y) = x^3 y^2 (1-x-y)$ .                                                                                      | 5 | L4 | 2 | 1 |
| (b) Find the polar coordinates of the following points:<br>(i) $(\sqrt{2}, \sqrt{2})$ (ii) $(-1, \sqrt{3})$ (iii) $(-\sqrt{3}, -1)$ (iv) $(1, -1)$ (v) $(3, 4)$ | 5 | L4 | 2 | 1 |

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|           |                                                                                                                                                                                       | Marks | BL | CO | PO |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|----|----|----|
| ✓ VI.     | (a) Change the order and hence evaluate $\int_0^1 \int_{x^2}^x (2x^2 + y) dy dx$ .                                                                                                    | 5     | L3 | 3  | 1  |
|           | (b) Evaluate the volume of the solid generated by revolving the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ about the $X$ axis.                                                   | 5     | L6 | 3  | 1  |
| <b>OR</b> |                                                                                                                                                                                       |       |    |    |    |
| VII.      | (a) Evaluate the arc length of one arc of the cycloid $x = a(\theta - \sin \theta)$ , $y = a(1 - \cos \theta)$ .                                                                      | 5     | L5 | 3  | 1  |
|           | (b) Find the volume of the sphere $x^2 + y^2 + z^2 = a^2$ using triple integration.                                                                                                   | 5     | L6 | 3  | 1  |
| <b>OR</b> |                                                                                                                                                                                       |       |    |    |    |
| VIII.     | (a) Show that $\vec{F} = (6xy + z^3)\hat{i} + (3x^2 - z)\hat{j} + (3xz^2 - y)\hat{k}$ is irrotational and find the potential function of $\vec{F}$ .                                  | 5     | L3 | 4  | 1  |
|           | (b) If $r =  \vec{r} $ , where $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ . Prove that $\nabla r^n = nr^{n-2}\vec{r}$ .                                                               | 5     | L5 | 4  | 1  |
| <b>OR</b> |                                                                                                                                                                                       |       |    |    |    |
| /IX.      | (a) Evaluate $\int_C \vec{F} \cdot d\vec{r}$ if $\vec{F} = 3xy\hat{i} - y^2\hat{j}$ where $C$ is the curve in the $XY$ plane given by $y = x^3$ from $(0,0)$ to $(1,1)$ .             | 5     | L4 | 4  | 1  |
|           | (b) Evaluate using Stoke's theorem $\oint_C \vec{F} \cdot d\vec{r}$ where $C$ is a curve in $XY$ plane around the lines $x=0, y=0, x=5, y=5$ and $\vec{F} = x^2\hat{i} + xy\hat{j}$ . | 5     | L4 | 4  | 1  |

Bloom's Taxonomy Levels

L1 = 14.3%, L2 = 9.5%, L3 = 19%, L4 = 23.9%, L5 = 19%, L6 = 14.3%.

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